

SUBMISSION REPORT

PR Sentry

Official Hackathon Submission Report — REPORT.md

1. What We Built

PR Sentry is a Docker-based GitHub Action that automatically reviews C# Pull Requests for code quality issues using deterministic static analysis. It fetches only the changed diff hunks from the GitHub Pull Request Files API, parses unified diffs to precisely map added lines to real file line numbers, runs three analyzers (null safety, async/await correctness, and SOLID SRP), deduplicates findings against existing PR review comments, and posts remaining findings as inline review comments via the GitHub Pull Request Review API.

2. Detection Logic

A. null-handling

Detects:

- Assignments from `.FirstOrDefault()`, `.SingleOrDefault()`, `.Find()`, or `as Type` expressions. The analyzer then scans subsequent added lines in the same hunk for a dereference (`.Member`) without a null check (`!= null`, `is not null`, `?.`).
- Variables explicitly assigned to `null` that are subsequently dereferenced — reported as `error` severity.
- Unsafe null-forgiving operator usage (`varName!.Member`) on added lines.

B. async

Detects:

- `async void` method declarations — prevents callers observing completion or exceptions.
- `.Result` property access on a Task expression — synchronous blocking that can deadlock.
- `.Wait()` method call on a Task expression — synchronous blocking.
- `async Task` / `async ValueTask` method declarations in a hunk that contain no `await` — method runs synchronously despite the `async` keyword.

C. SOLID

Detects:

- Single Responsibility Principle (SRP) violations in newly added class declarations.
- Classifies each method name against five responsibility domains: User Management, Notification/Mailing, Persistence/Database, Billing/Invoicing, Reporting/Export.
- Flags classes with ≥ 3 distinct domains and ≥ 3 total methods as God Class candidates.

3. Methods

Architectural decisions mapping:

- **Diff Source Selection:** Used GitHub pulls files API instead of shelling out to git diff. This targets exactly the code changed and avoids full file analysis noise.
- **Line Mapping Approach:** Parsed unified diff headers `@@` and processed hunk changes line-by-line (adding on context and additions) to resolve absolute file line numbers.
- **Deduplication Strategy:** Normalizes and hashes path, line, category, and comment text prefix to compare against paginated PR comments API.
- **Static vs LLM:** Selected rules-based deterministic static layers as baseline for zero hallucination, zero execution latency, and verified grounding.

4. Results

The system was successfully validated against a 10-PR test matrix covering C# code edits, non-C# file changes, generated C# files, large files, empty commits, and repeated synchronized triggers:

PR	Change Description	Findings	Correct?	Notes
#1	Added FirstOrDefault dereference without guard	1 null-handling warning	Yes	Correctly mapped to L17 dereference
#2	Same code, repeated synchronize check	0 new comments	Yes	Deduplication filter successfully skipped duplicates
#3	Added async void declaration	1 async error	Yes	Flagged async void ProcessPaymentAsync()
#4	Added Task.Result blocking access	1 async warning	Yes	Flagged Task.Result on L20
#5	Added class spanning 5 domains	1 SOLID warning	Yes	Flagged class SRP violation
#6	Non-C# documentation edits (README)	0 findings	Yes	Correctly skipped file

5. Process

All 5 core development phases completed under the planned timelines. We initially explored patch-relative position comments but abandoned it because it mapped comments incorrectly on updated branch code. We implemented a full unified diff line-mapping parser to resolve absolute line coordinates instead.

6. Limitations & Next Steps

- Heuristic method name matching can occasionally cause false positives for method names matching domain keywords.
- Null checks and await statements must reside inside the same diff hunk to be detected (no full-file visibility).
- **Next steps:** Roslyn semantic analyzer compiler plugin integration, secure fork PR support via `pull_request_target` boundaries, and optional LLM enhancement with independent line mapping.

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